SENG 696

A Diplomacy-Style Negotiation Game with Autonomous Agents

MaSE Analysis and Design

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October 7, 2026

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1 Introduction

This document is a Multiagent Systems Engineering (MaSE) rundown of the Diplomacy-style multi-agent system (MAS) video game, in which 1 human player will compete with several AI agent players to conquer the map and its territories. This document is broken down into an analysis phase and a design phase. In this, it will define the 4 agent classes and cover all their interactions and parts.

2 System Architecture Overview

The system is built from four agent classes. The **Player Interface** is the human player's agent. It shows the board and a negotiation inbox in a GUI and turns the human's choices into messages. An **AI Player** controls one autonomous power. It holds private beliefs about the board, a trust score for every other nation, a record of commitments and a personality profile fixed at the start of the game, and from these it decides what to propose, what to accept and which orders to give. The **Game Master** runs the game. It creates the map and then repeats the turn cycle: it opens a negotiation phase, then an order phase in which every unit receives exactly one order, resolves all orders at once, and broadcasts the result. The **Logger** records the accepted agreements and the board and results of every turn, reports after each turn which agreements were kept and which were broken, and exports a recap when the game ends.

Powers negotiate directly with one another: the Player Interface and the AI Players exchange private messages peer to peer, and the Game Master takes no part in them. Every power talks to the Game Master to receive phase announcements and turn results and to submit its orders. Every power reports the agreements it accepts to the Logger, and the Game Master sends the Logger the board and results of each turn. The Logger in turn sends relationship updates to every power and the final recap to the Player Interface. The Game Master holds the only authoritative board. Each power keeps a copy that it updates only from the Game Master's broadcasts, so no power can change the game state, and no power sees another power's orders until the Game Master publishes the turn result. Figure 14 in Section 4.1 shows the four classes and their conversations.

3 Analysis Phase

3.1 Goal Hierarchy

Figure 1 shows the goal hierarchy derived from the functional requirements, and the numbered list below it restates every goal.

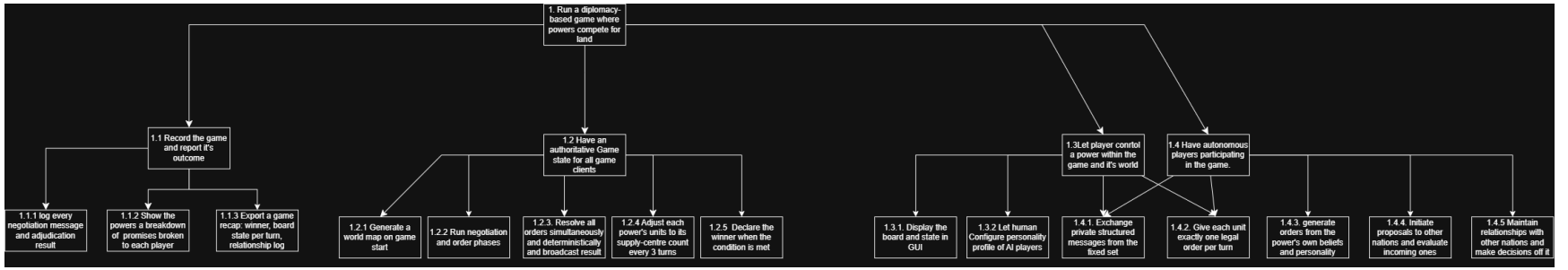


Figure 1: Goal hierarchy diagram

1 Run a diplomacy-based game where powers compete for land

- **1.1** Record the game and report its outcome
  - **1.1.1** Log every negotiation message and adjudication result
  - **1.1.2** Show the powers a breakdown of promises broken to each player
  - **1.1.3** Export a game recap: winner, board state per turn, relationship log
- **1.2** Have an authoritative game state for all game clients
  - **1.2.1** Generate a world map on game start
  - **1.2.2** Run negotiation and order phases
  - **1.2.3** Resolve all orders simultaneously and deterministically and broadcast the result
  - **1.2.4** Adjust each power's units to its supply-centre count every 3 turns
  - **1.2.5** Declare the winner when the condition is met
- **1.3** Let the player control a power within the game and its world
  - **1.3.1** Display the board and state in the GUI
  - **1.3.2** Let the human configure the personality profile of AI players
  - **1.4.1** and **1.4.2**, shared with goal 1.4 and listed there
- **1.4** Have autonomous players participating in the game
  - **1.4.1** Exchange private structured messages from the fixed set
  - **1.4.2** Give each unit exactly one legal order per turn
  - **1.4.3** Generate orders from the power's own beliefs and personality
  - **1.4.4** Initiate proposals to other nations and evaluate incoming ones
  - **1.4.5** Maintain relationships with other nations and make decisions based on them

Goals 1.4.1 and 1.4.2 are shared between 1.3 and 1.4 because the specification requires the human player to negotiate and give orders under the same rules as the autonomous powers, so both parents need them.

3.2 Use Cases and Sequence Diagrams

Four use cases cover a complete game from start to finish. Each is refined into a sequence diagram whose lifelines are roles. In the negotiation diagram, Power A and Power B stand for any two powers, human or autonomous.

3.2.1 Start Game

This use case appears when a player chooses to configure all AI powers and start a game with them. Once the Game Master has received it, assigned the AI players their powers, generated a world and updated the Logger with the first board state, the player and the AIs will all receive the board to begin playing the game.



Figure 2: Sequence diagram: Start Game

3.2.2 Negotiation Phase

This use case will have the Game Master kick off the negotiation phase by broadcasting its start to all players. At this point a player may choose to send an agreement to another player. The responding player can either accept or reject the offer, leaving relationship effects either way. If an agreement is struck and accepted, then the Logger will be notified of the successful deal. There can be many different negotiations between many different players all at the same time. To break the loop, the human player must click the End Negotiations button to stop it and transition to the order phase of the game.

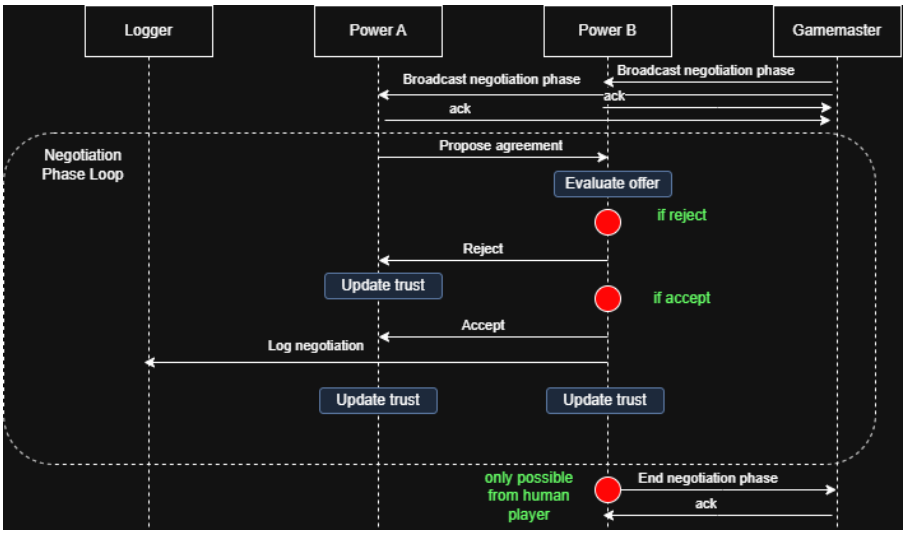


Figure 3: Sequence diagram: Negotiation Phase

3.2.3 Adjudicate Orders

This use case appears every game cycle and happens when the transition from the negotiation phase to the order phase is triggered. Each player participating in the game will give a list of orders for all their units to the Game Master. The Game Master, once it has collected them all, will resolve all orders at the same time and then update the players' unit counts to reflect the supply centre control every 3 turns. Then the new board state is logged, and all breached relationships and agreements are recorded and sent to the respective participating players. Then the Game Master decides if the game is finished or should continue based on the win condition. If the game is not over then it will loop back to the negotiation phase.

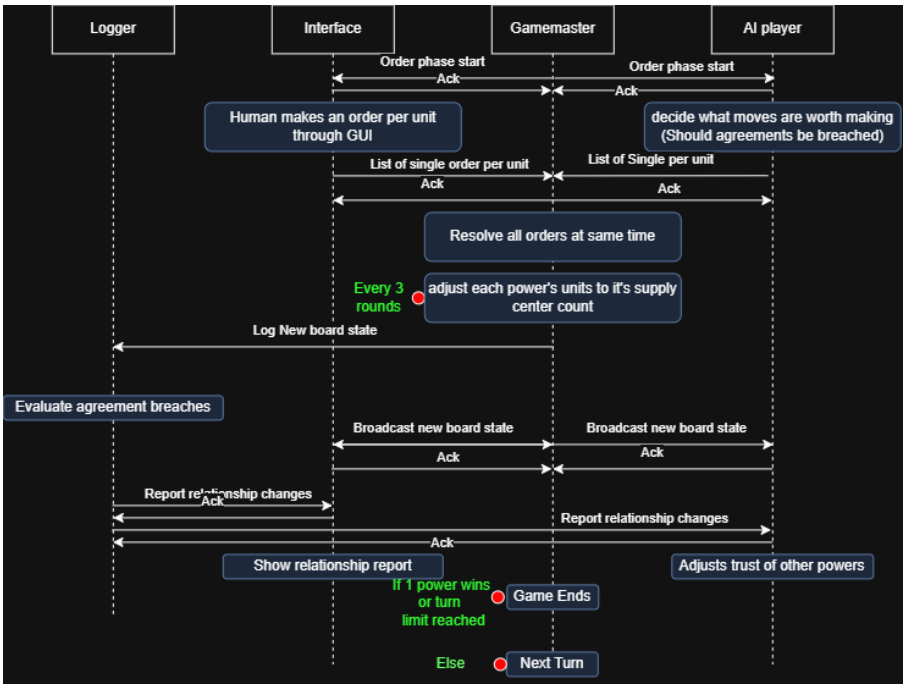


Figure 4: Sequence diagram: Adjudicate Orders

3.2.4 End of Game

End of Game is a use case that will appear once per game and be the signal for all elements to return to their original dead states. This will also trigger the Logger to send a full recap to the Interface for the player to look at the prior match and a breakdown of what happened.

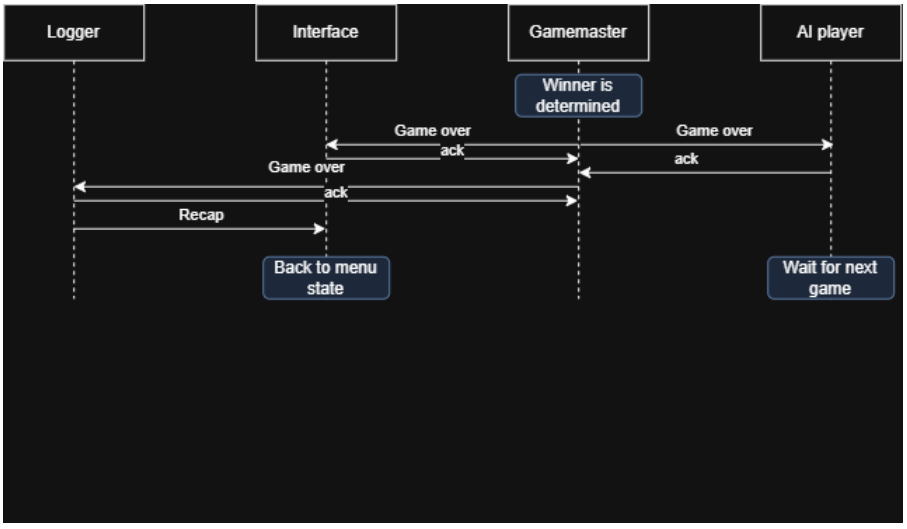


Figure 5: Sequence diagram: End of Game

3.3 Role Model

Figure 6 shows the role model. Each role box lists the goals assigned to the role and is attached to its tasks, drawn as ovals. Solid arrows are communication between tasks of different roles, and

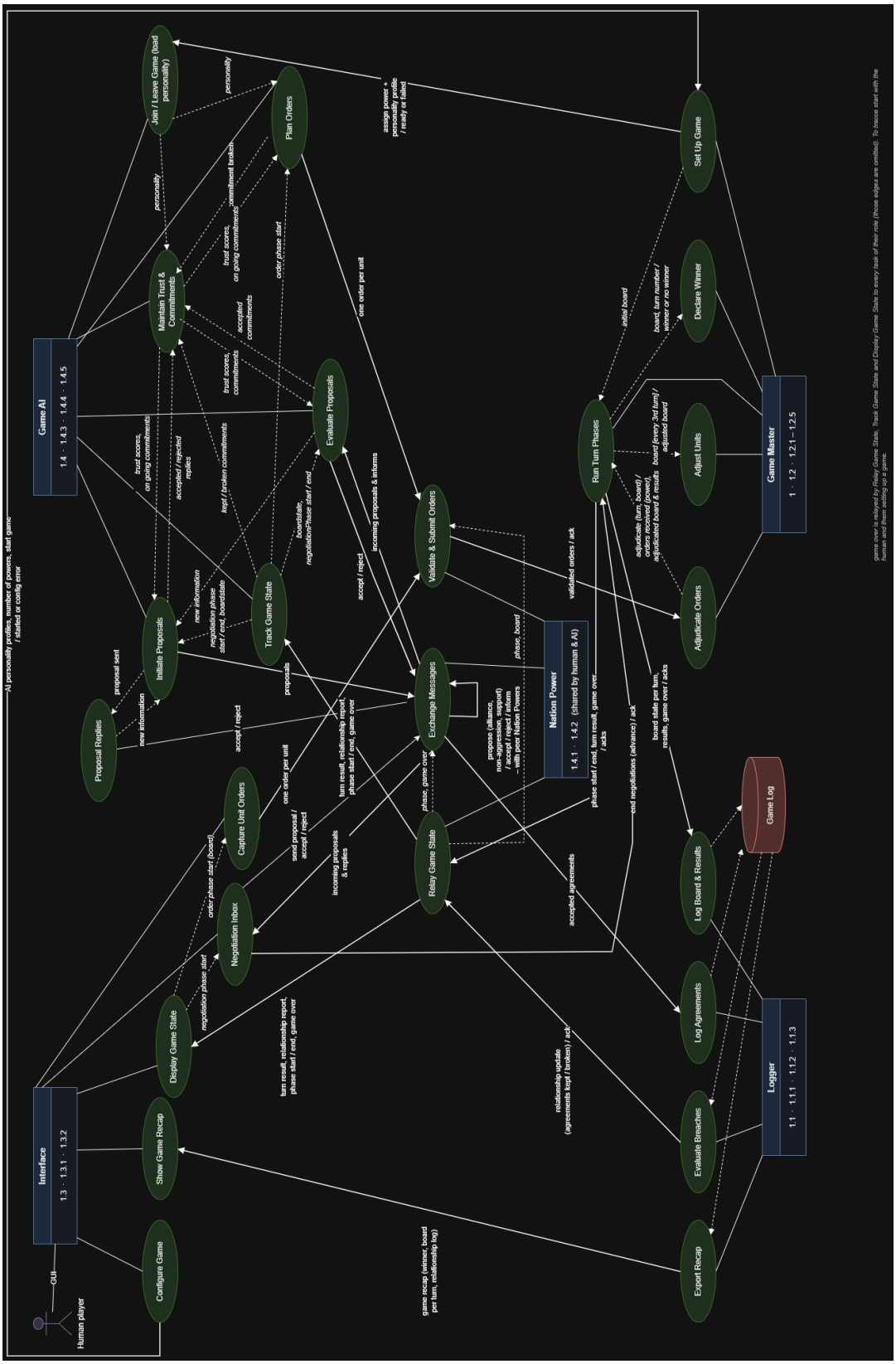


Figure 6: MaSE role model with tasks

dashed arrows are communication between tasks of the same role.

Interface. This role is the human player's window on the game. It will allow them to configure, start, view and play the game. It will also give a summary of each turn and of the game in total. This interface will also help direct the turns of the game.

Nation Power. The Nation Power role is here to normalize the interactions between two nation powers. These powers are either AI or human, and this role is here to allow them to communicate and participate in the game in a way that adheres to the rules and phases of the game. This means negotiating between powers and ordering units on the board. It is also in charge of delivering each turn's updates to the respective player in a way that it will understand.

Game AI. The Game AI role is there to allow AI players to interact and play the game with some logical systems behind the scenes. It will allow them to be given a personality described by the player and then make decisions based on that personality when given a board state. It will fill the role of brain for these nation powers and give them reasoning and tracking of the world around them.

Game Master. The Game Master role is here to start the game, generate the world, dictate turns and orders, and streamline all turns of the game, as well as to declare a winner.

Logger. The Logger has the role of timekeeping each round of the game, as well as tracking the agreements that two nations have committed to and the state of each agreement at different stages of the game. It is also responsible for updating the human player with a recap when the game is deemed over.

3.4 Concurrent Task Diagrams

Concurrent task diagrams are given for the seven tasks that carry the coordination and decision logic of the system, grouped by role. The diagrams for the remaining tasks were excluded in the interest of space.

3.4.1 Interface

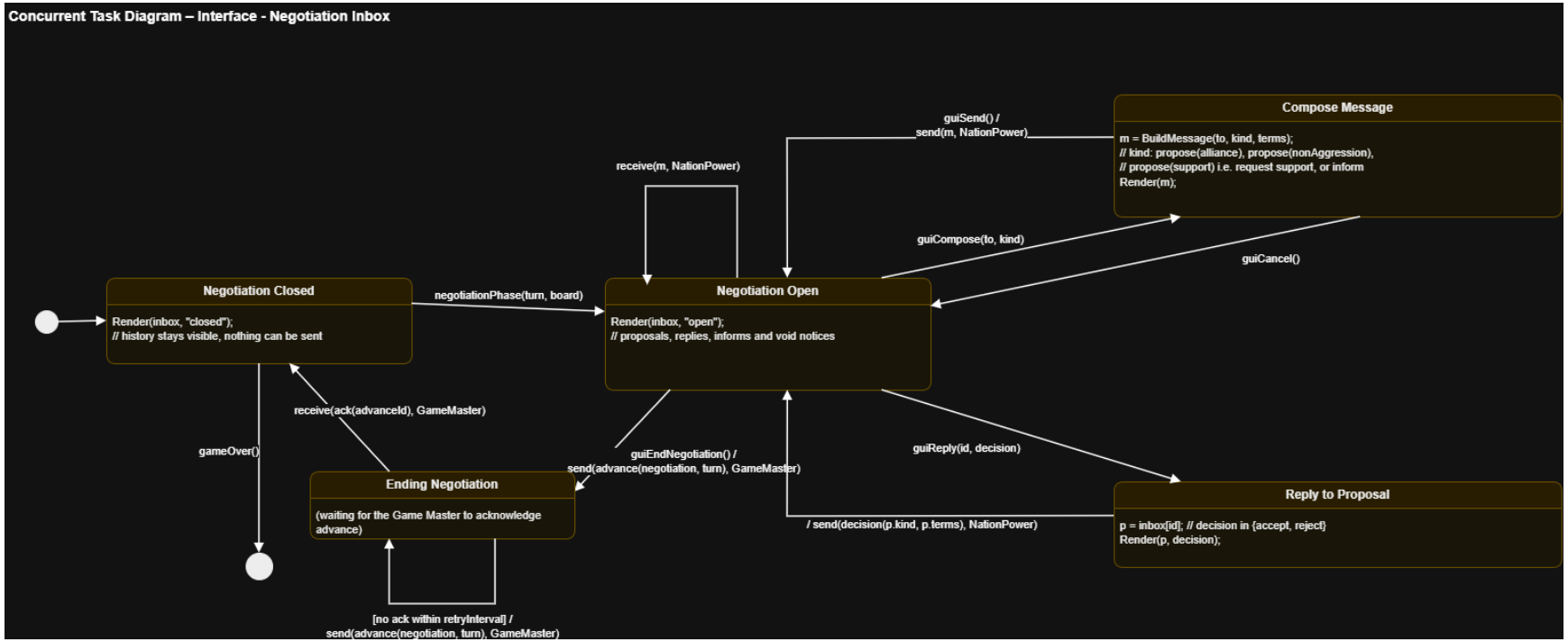


Figure 7: Concurrent task diagram: Negotiation Inbox

3.4.2 Nation Power

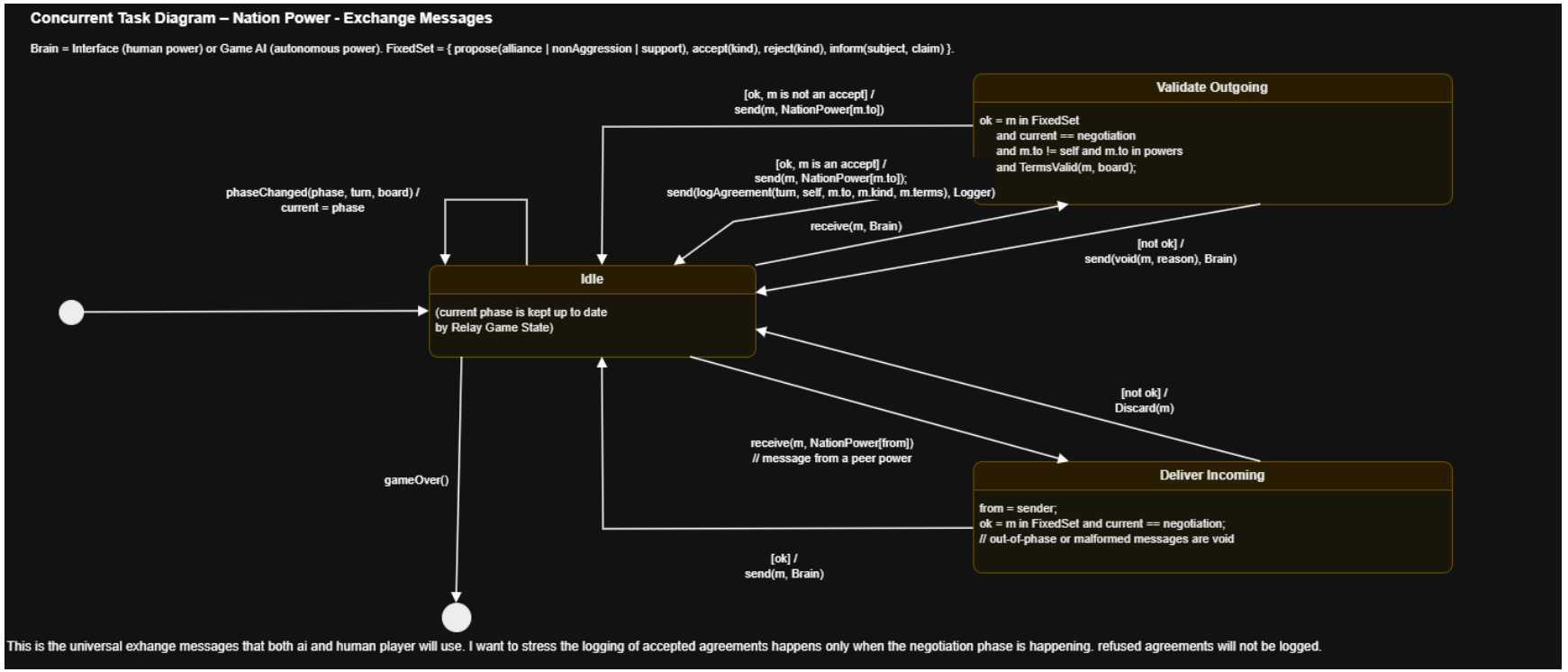


Figure 8: Concurrent task diagram: Exchange Messages

3.4.3 Game AI

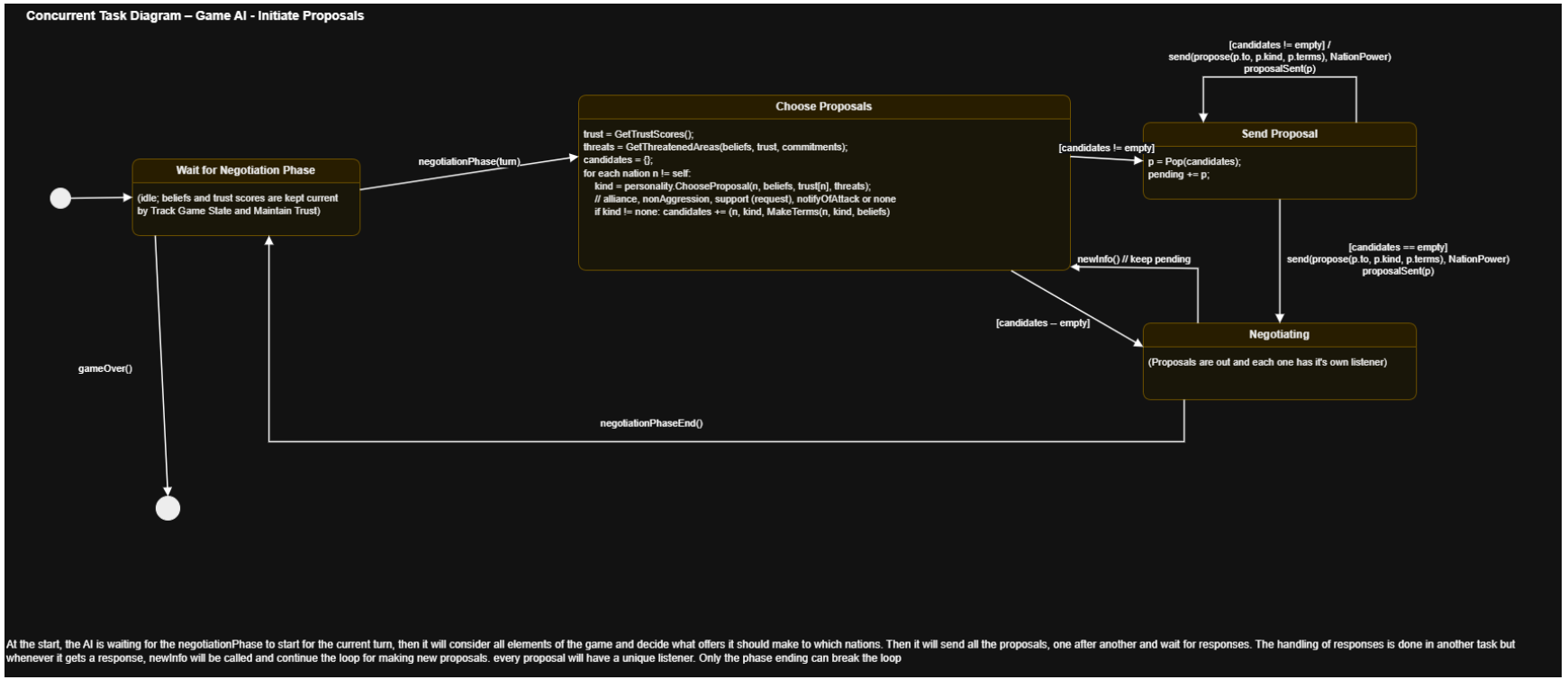


Figure 9: Concurrent task diagram: Initiate Proposals

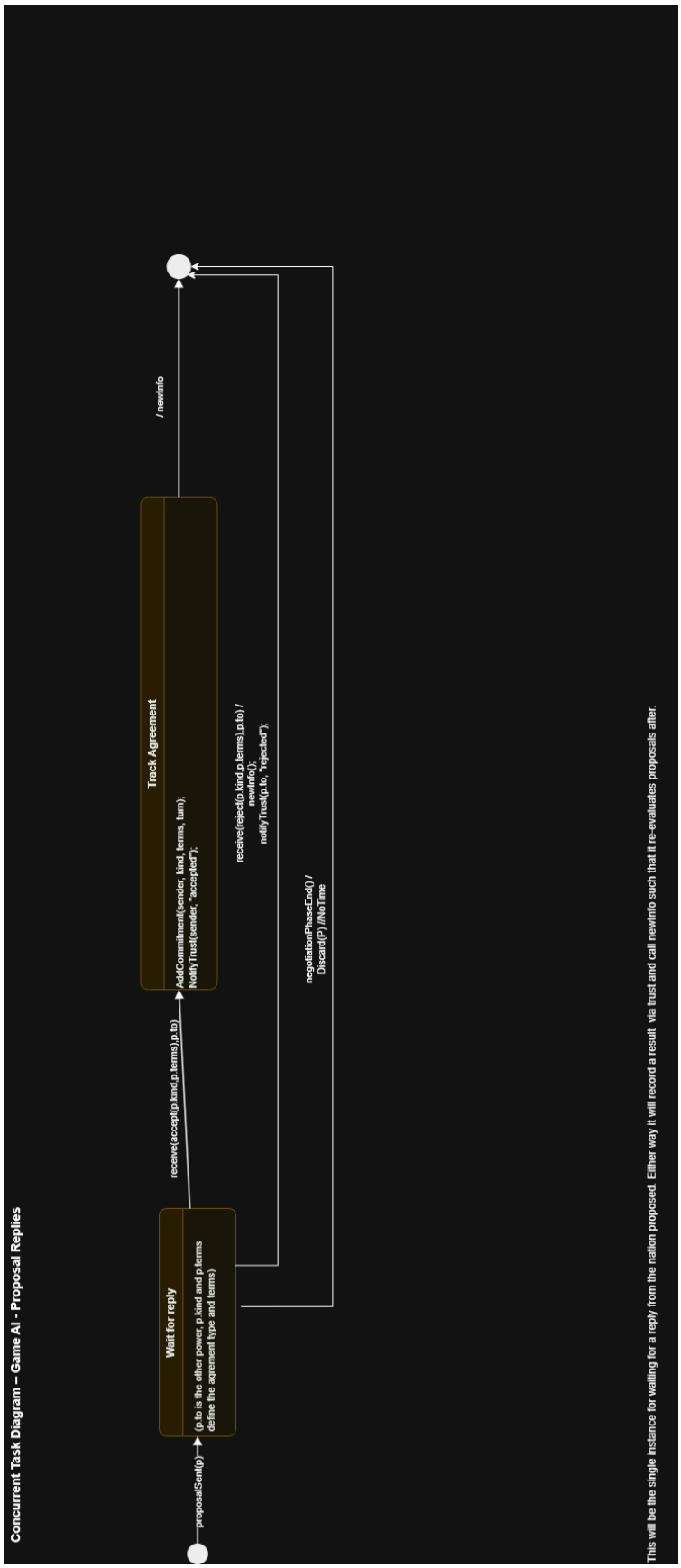


Figure 10: Concurrent task diagram: Proposal Replies

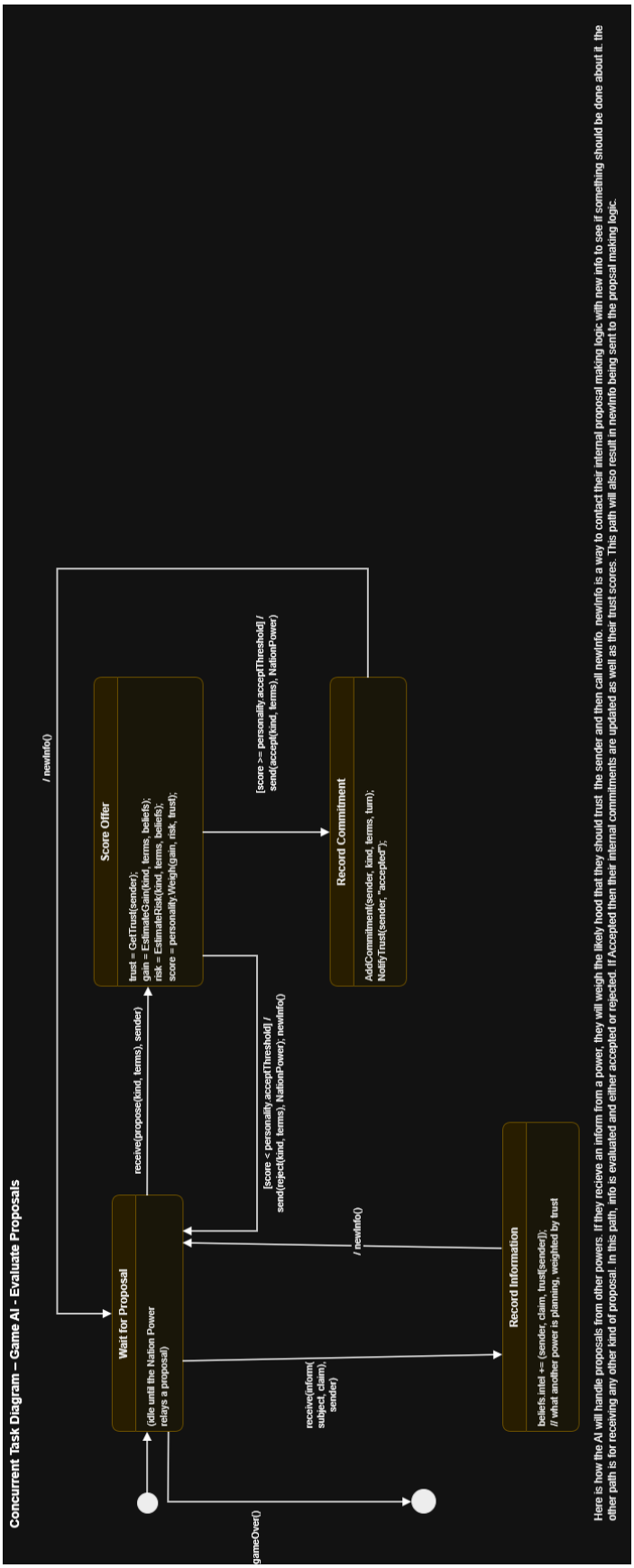


Figure 11: Concurrent task diagram: Evaluate Proposals

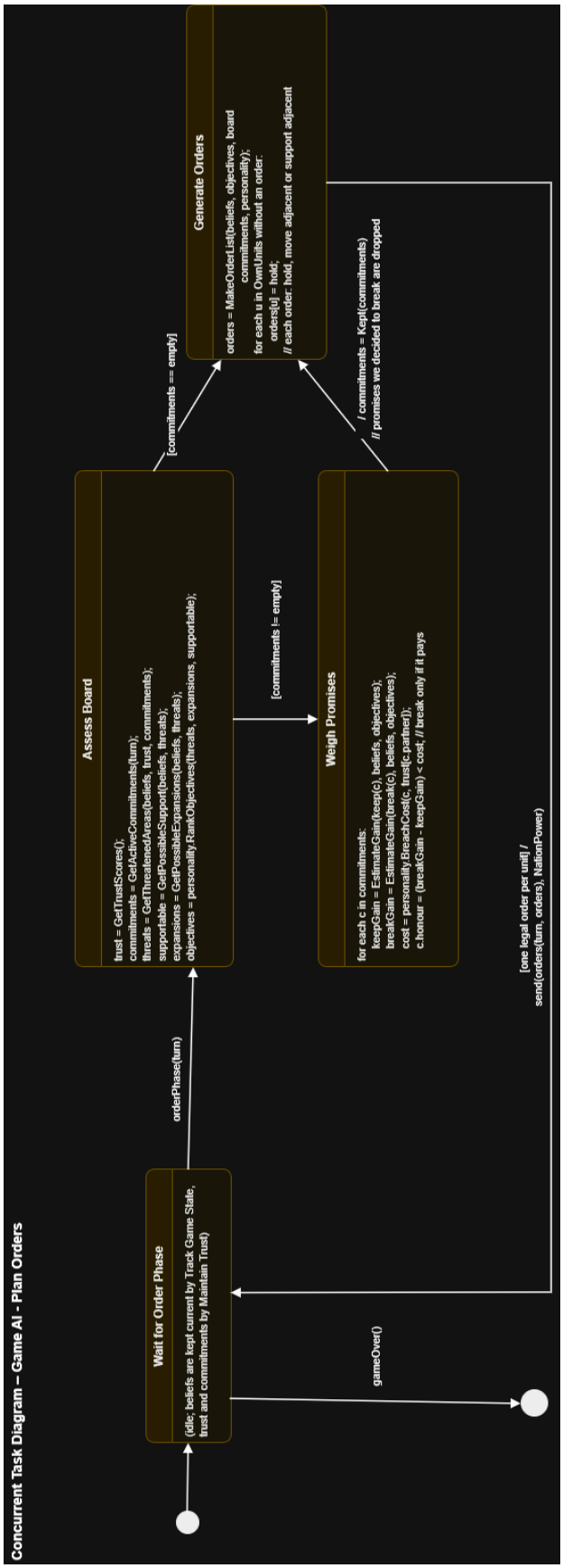


Figure 12: Concurrent task diagram: Plan Orders

3.4.4 Game Master

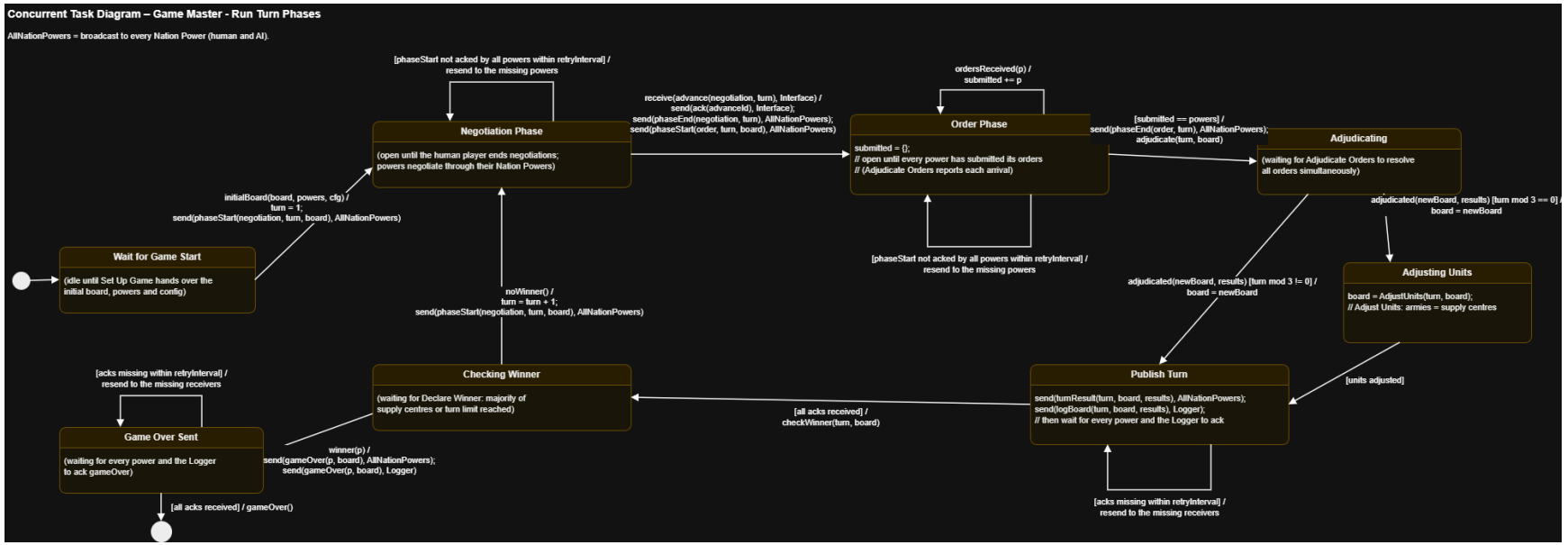


Figure 13: Concurrent task diagram: Run Turn Phases

4 Design Phase

4.1 Agent Classes

Figure 14 shows the four agent classes, their roles and the conversations between them. A description of each agent follows.

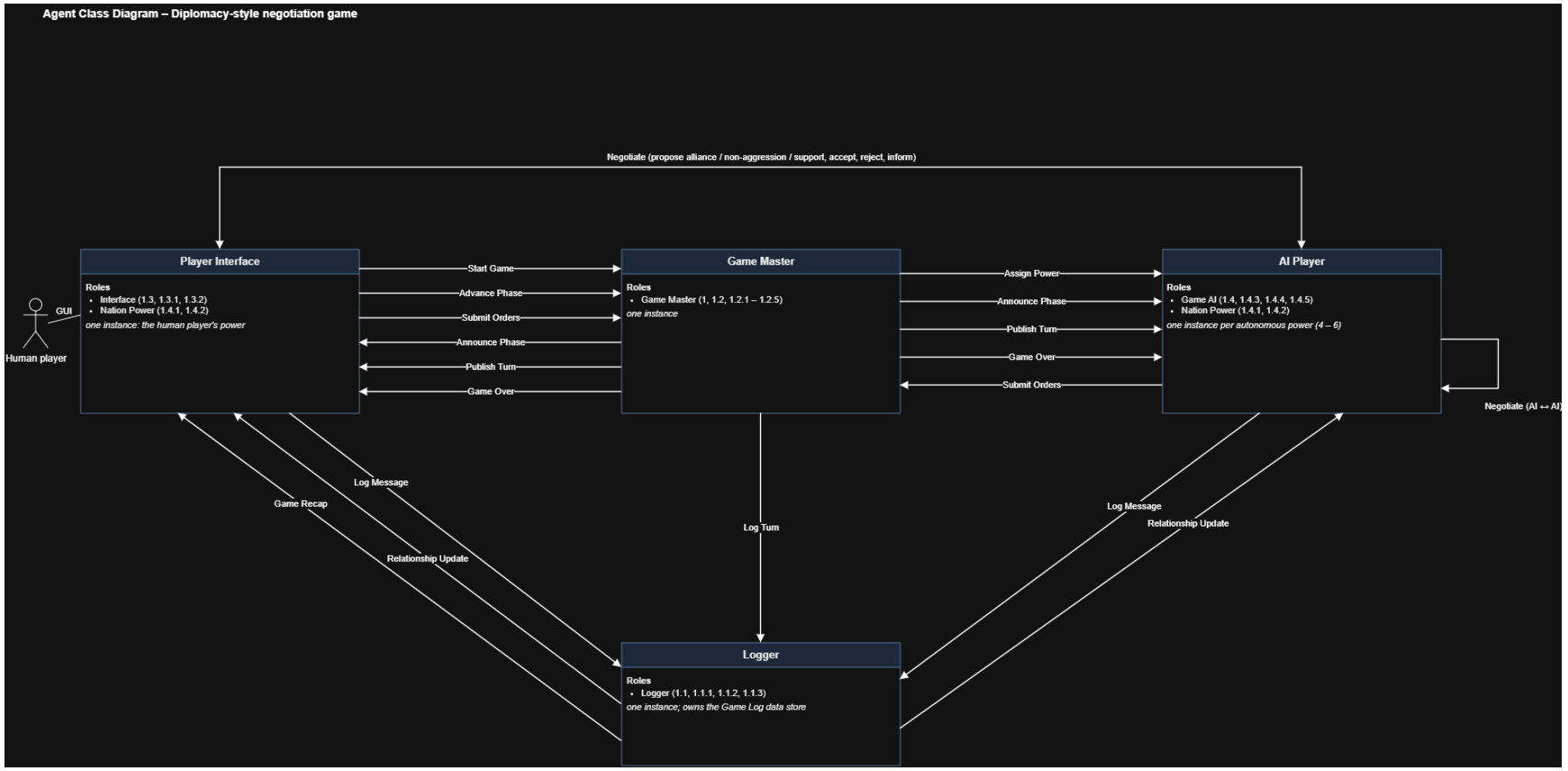


Figure 14: Agent class diagram

4.1.1 Player Interface

The Player Interface is the agent that acts as a view into the game. It allows the player to set up the game and to interact with the world and the other nation powers. It will show the board, units, territories and supply centres. The view will be updated every turn to show the most current info and allow the player to remain knowledgeable of the current game. It will allow them to order units within the rules of the game and communicate with other powers. This view is also in charge of determining when the negotiation phase is complete. After each turn it also shows which of the player's agreements were kept or broken. When the game is over it will display a recap.

It is a separate agent because it is a way for a human player to partake in the game while being bound by the same rules and restrictions as the AI players. This allows for universal communication between the AI powers and the player. It must be separate from the Game Master and the Logger as it must only know what the human player's power should know and not have access to any type of authority over the world. There is one Player Interface, for the human player.

4.1.2 Game Master

The Game Master is the authoritative force in adjudication of the board and its state, as well as maintaining and directing the game phases. It drives the game forward and is a source of guidance for all other agents. It will update all of them on changes such as victory. At the start it generates the map and assigns each AI Player its power and personality. Each turn it collects every power's orders and resolves them all at once.

It is separate because it must maintain an authoritative game state that no other part can affect, and it is a neutral party that all can rely on as a source of truth. There is one Game Master per game.

4.1.3 AI Player

AI Players must be capable of playing the game without a human interacting with them, acting in their own interests and partaking in the game while adhering to all the same rules as the others. They must be capable of partaking in negotiations and ordering units all on their own while not breaking any of the restrictions on knowledge that the game enforces. Each one keeps its own beliefs about the board, a trust score for every other power and a record of its commitments, and a personality set at the start of the game shapes its decisions.

It must be separate because each power is an autonomous, self-interested power that tracks all elements of the game. Each one must do this and share it with no one else. This means the Player Interface cannot know what they are doing, and neither the Game Master nor the Logger can share knowledge with the Game AI. There is one AI Player for each autonomous power.

4.1.4 Logger

The Logger is in charge of recording the game and all agreements and promises. It is also in charge of tracking the relationships between powers and reporting them to the powers privately. After each turn it checks the results against the logged agreements to find which were kept or broken, and at the end of the game it exports a recap.

It is separate as it lowers the load on the Game Master of tracking all the elements and relationships at the same time. It is separate from the game powers as it is important to keep their knowledge at an appropriate level such that it doesn't break the restrictions of the game. There is one Logger per game.

4.2 Conversations

Twelve conversations connect the agent classes. Figures 15 to 18 give the communication class diagrams for the initiator and the responder of each one.

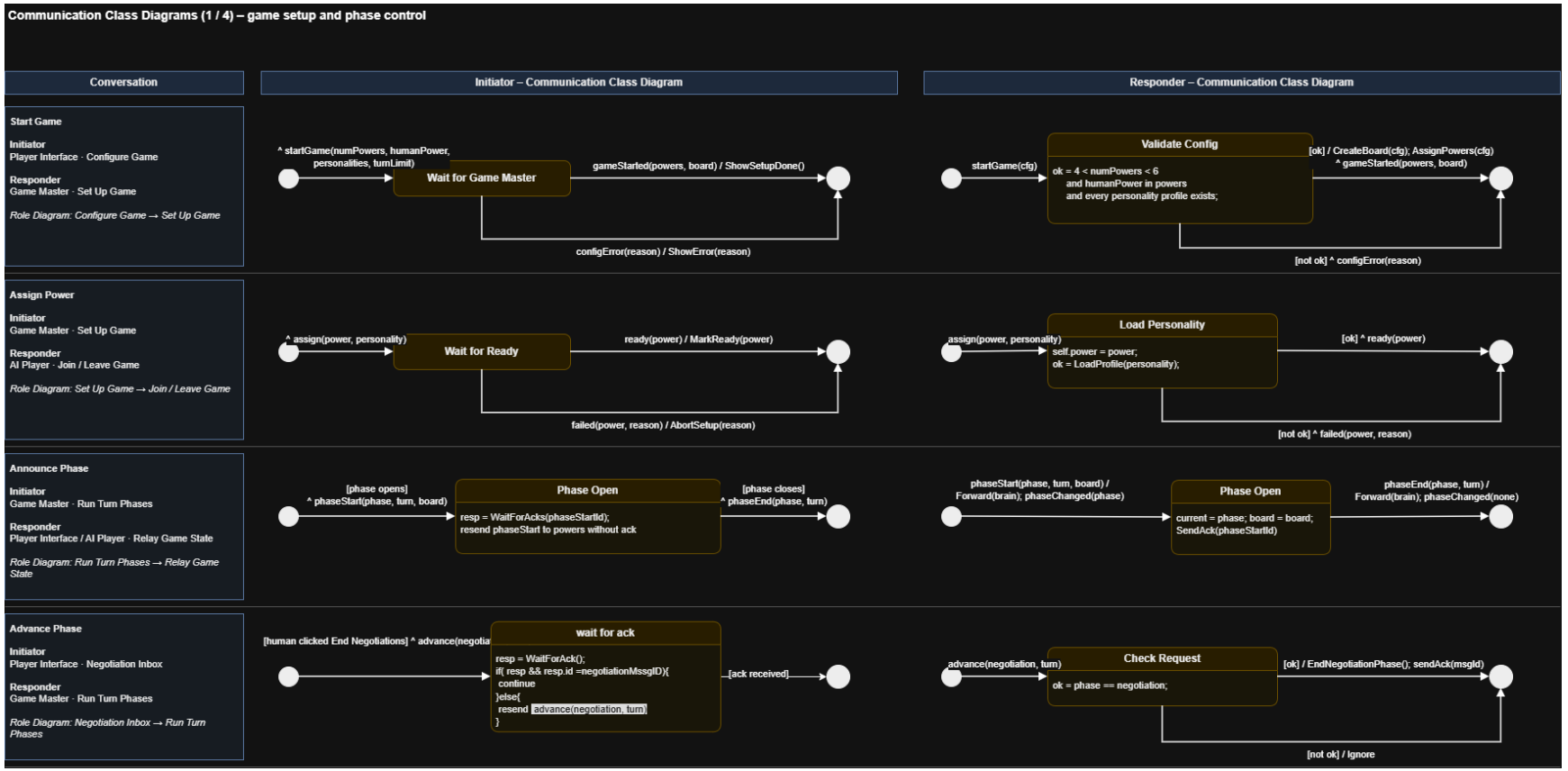


Figure 15: Communication class diagrams: Start Game, Assign Power, Announce Phase and Advance Phase

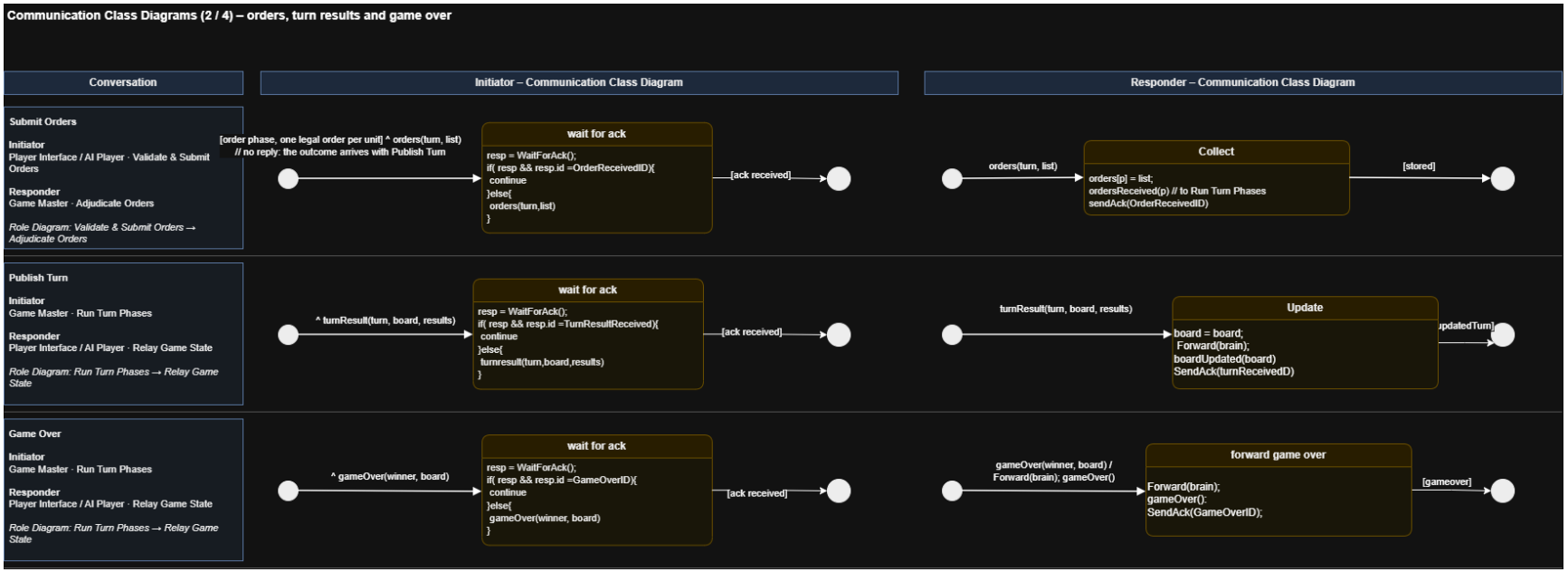


Figure 16: Communication class diagrams: Submit Orders, Publish Turn and Game Over

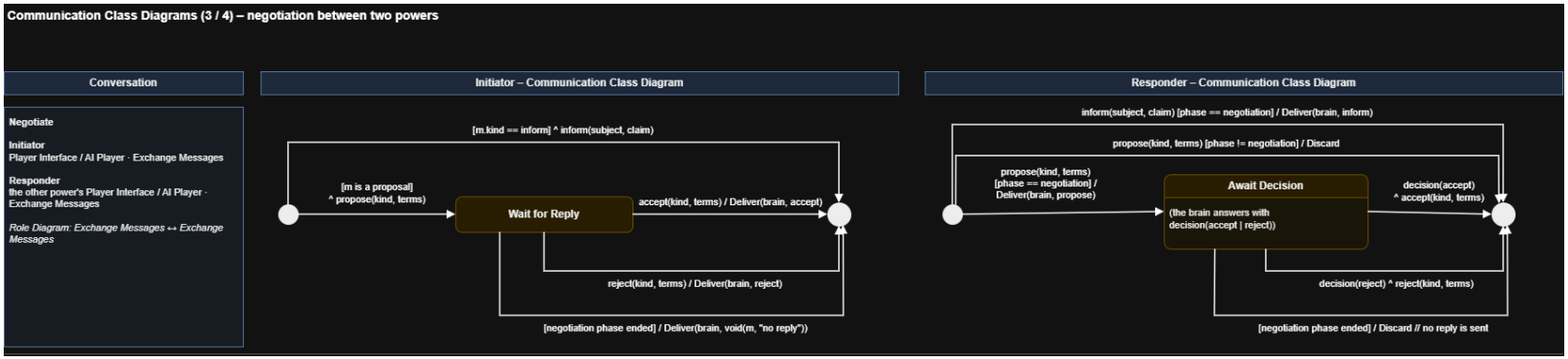


Figure 17: Communication class diagram: Negotiate

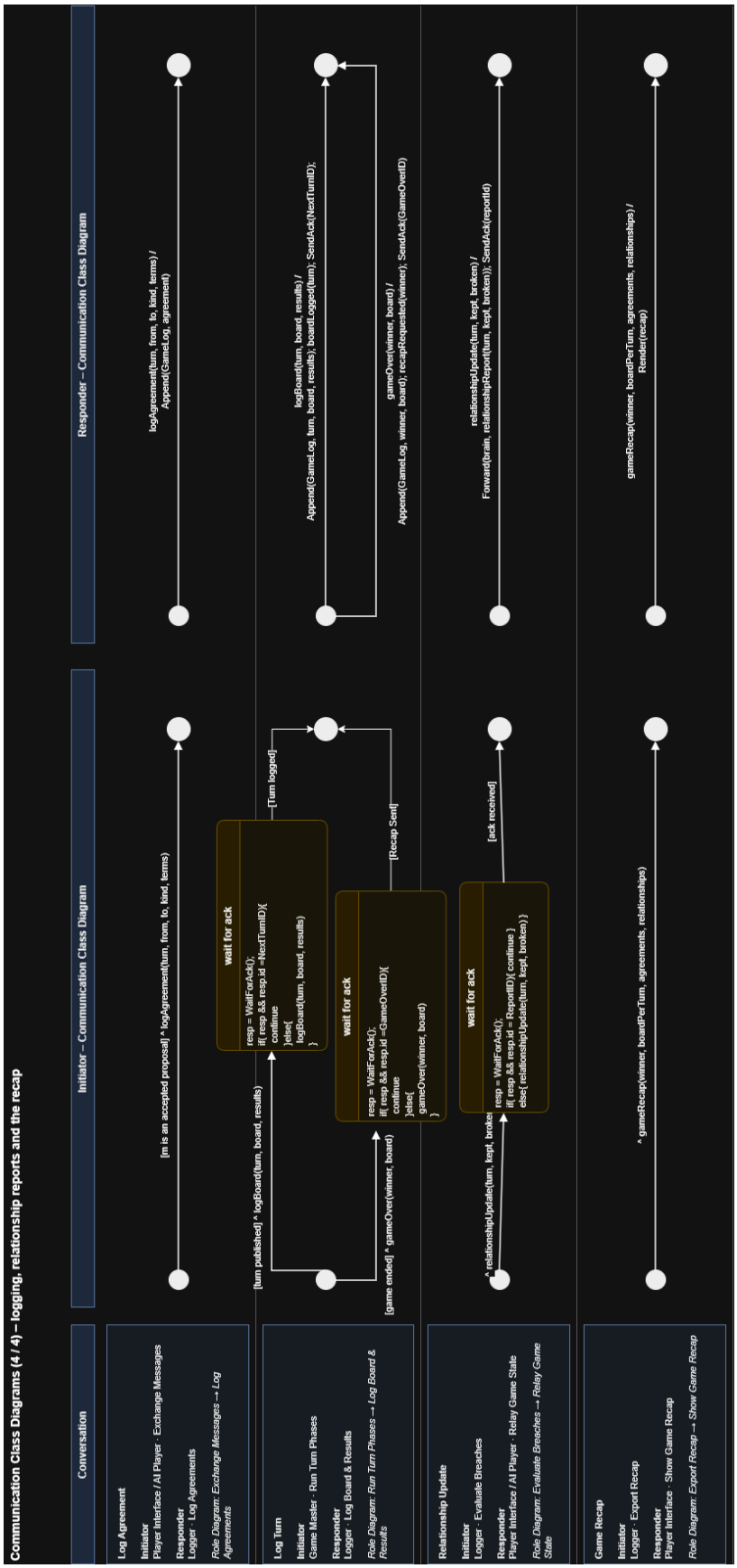


Figure 18: Communication class diagrams: Log Agreement, Log Turn, Relationship Update and Game Recap

4.3 Agent Architectures

Each agent class is assembled from components that correspond to the tasks of the roles it plays, and dashed arrows are conversations with other agents. The three Nation Power components are one implementation that is included in both the Player Interface and the AI Player.

4.3.1 Player Interface

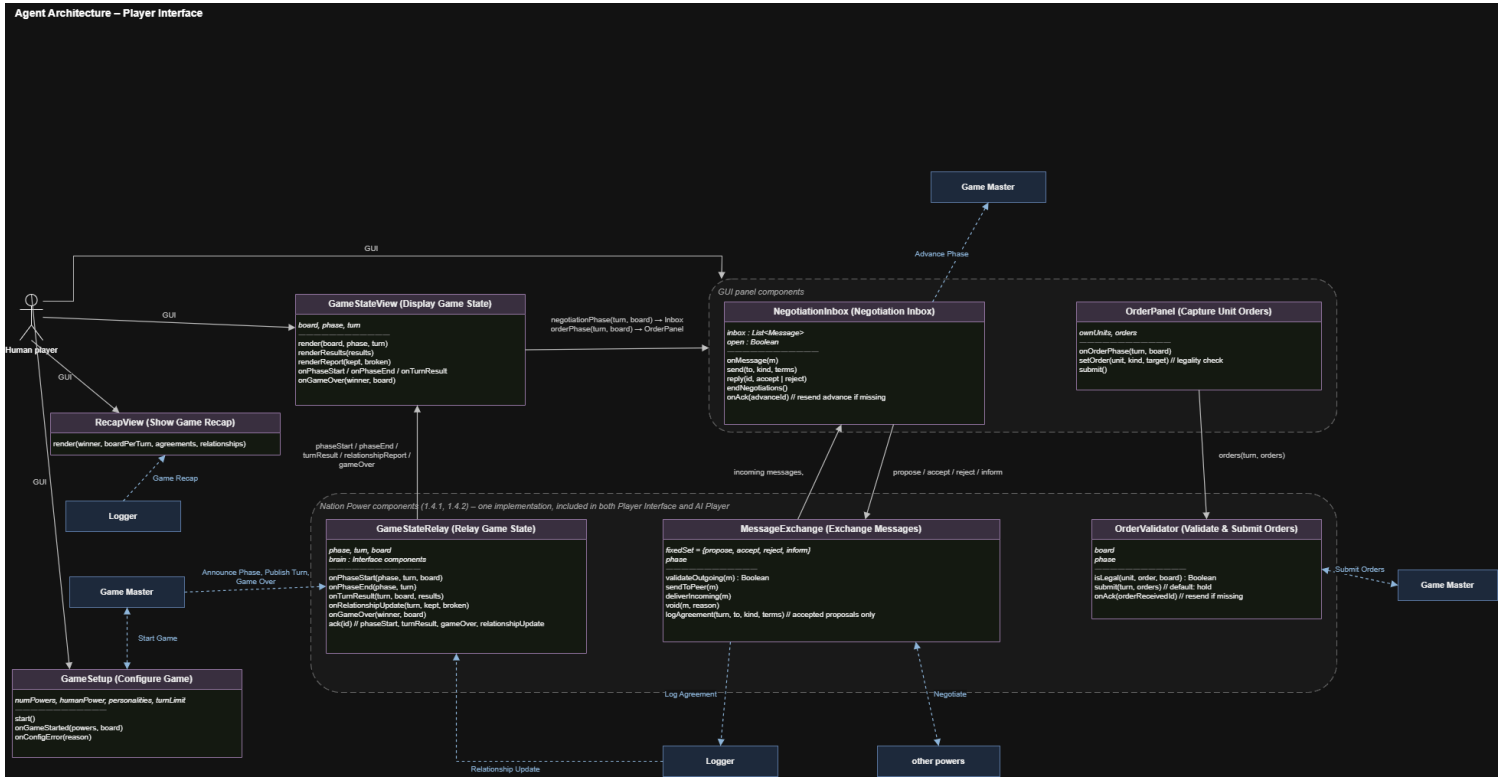


Figure 19: Agent architecture: Player Interface

4.3.2 Game Master

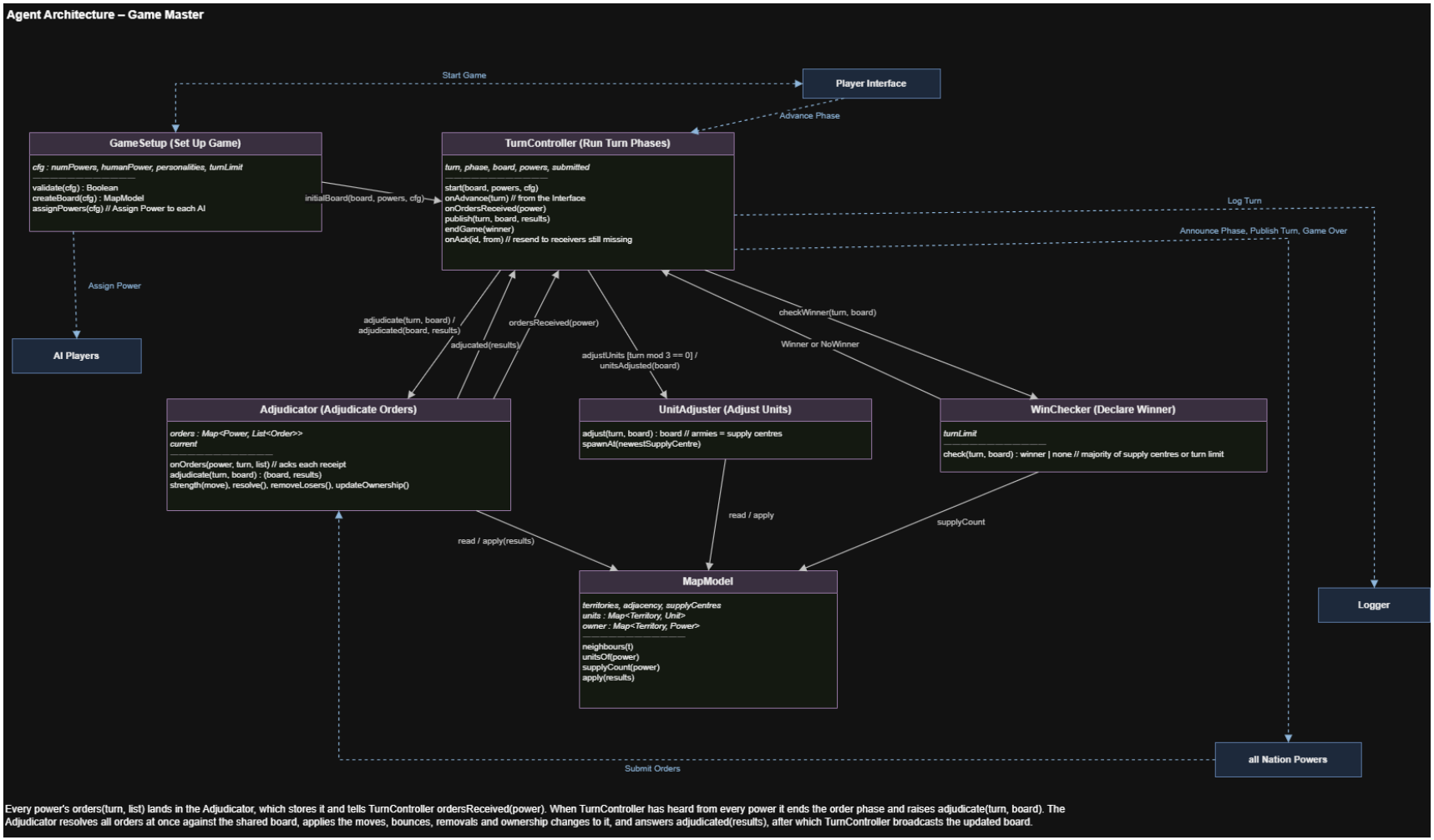


Figure 20: Agent architecture: Game Master

4.3.3 AI Player

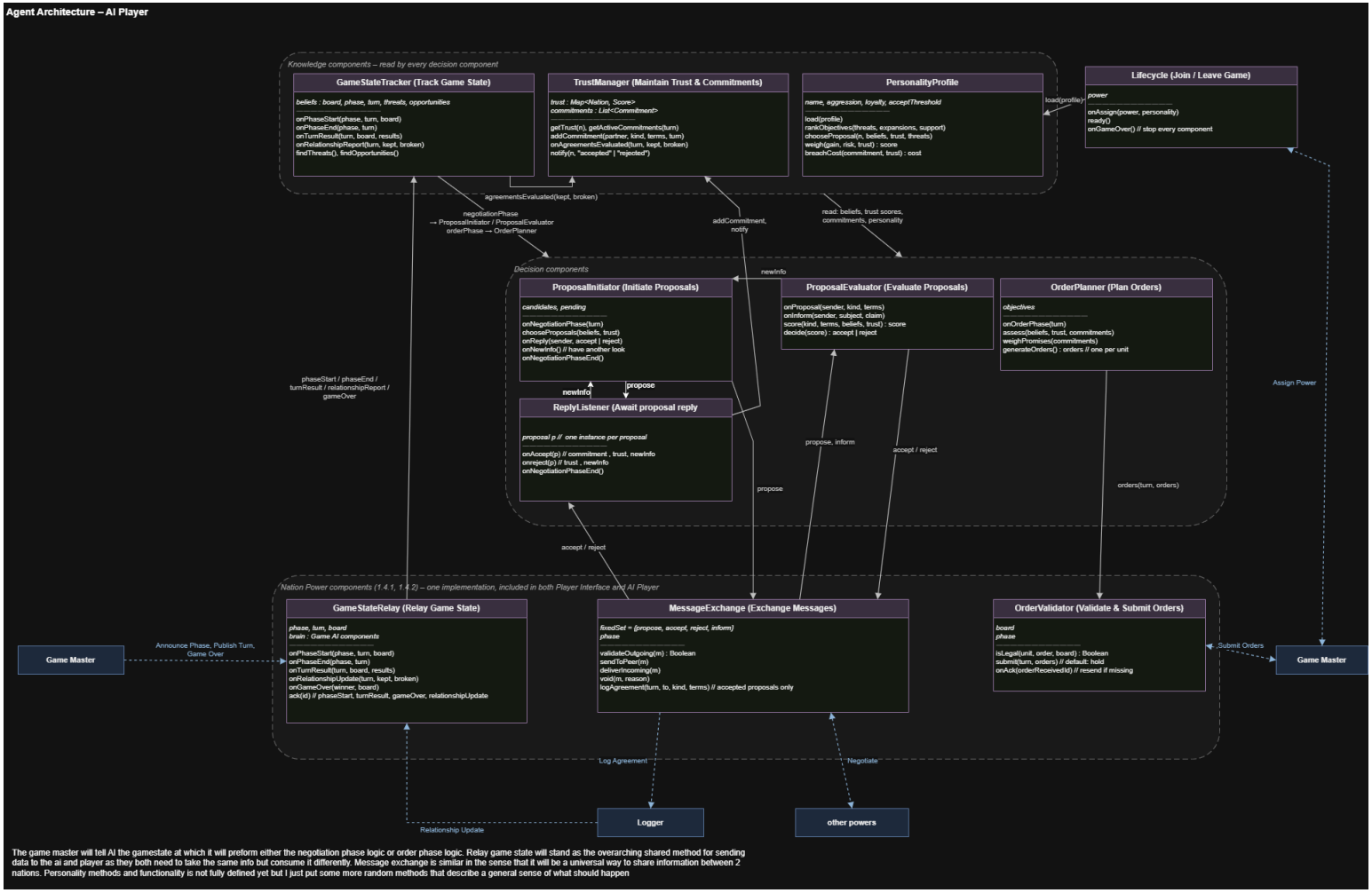


Figure 21: Agent architecture: AI Player

4.3.4 Logger

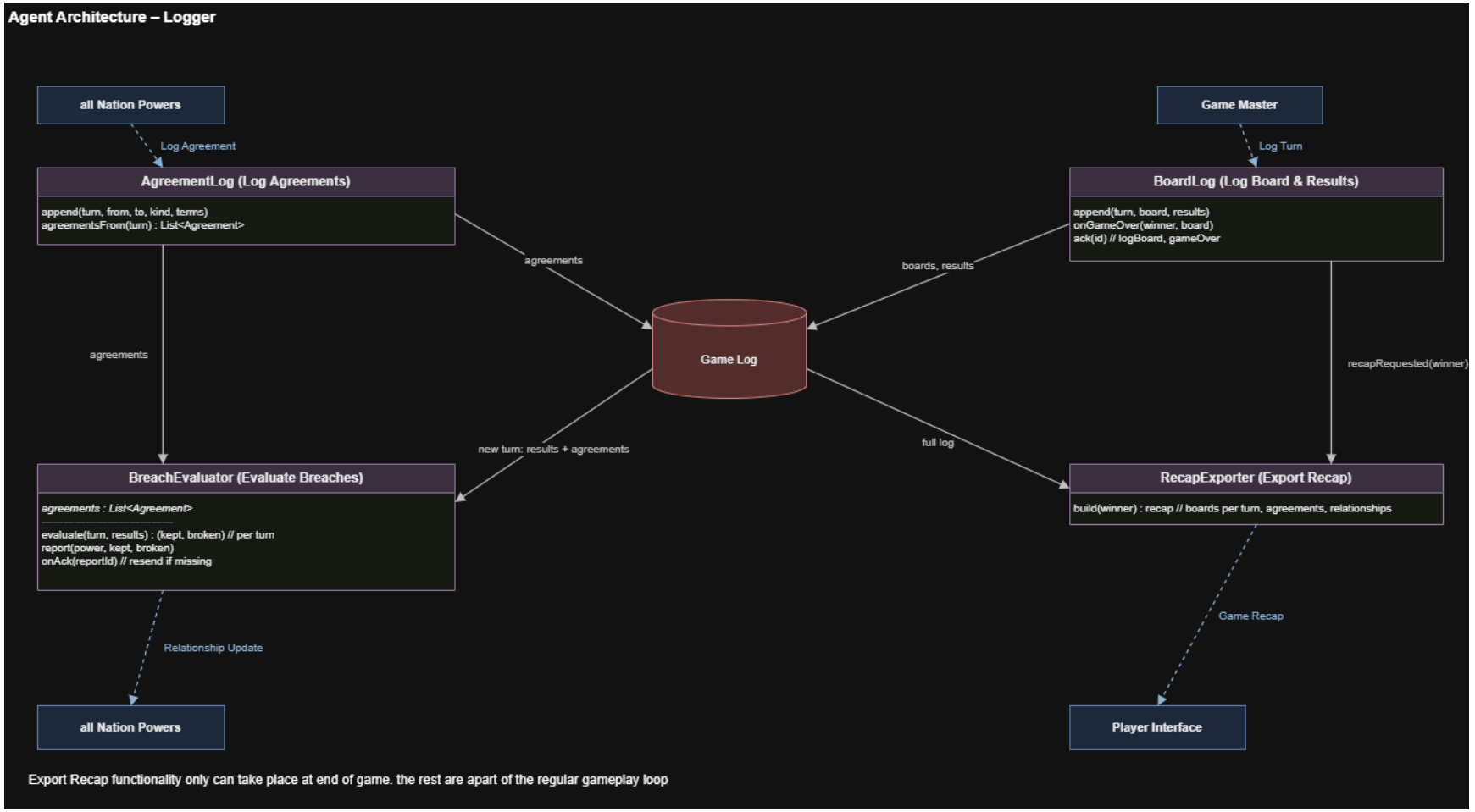


Figure 22: Agent architecture: Logger

4.4 Deployment

Figure 23 shows the deployment. There is one Player Interface, on the human player's machine. The Game Master and the Logger, with its Game Log, run together on the game server, one of each per game. There is one AI Player per autonomous power.

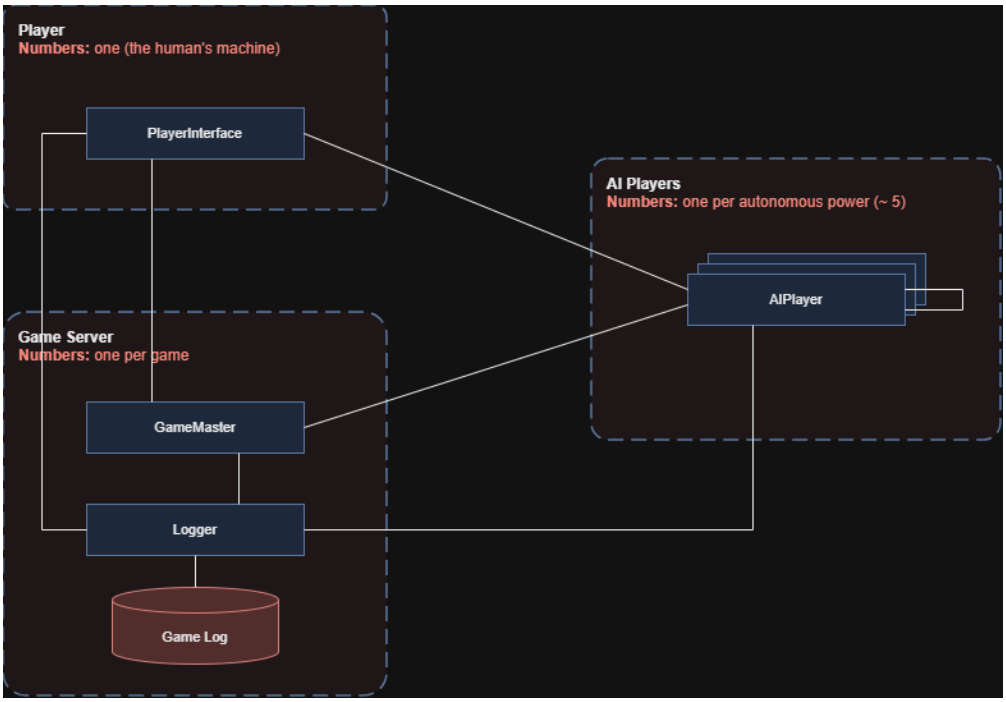


Figure 23: Deployment diagram